Steering a Network with Its Own Reflection: True, False, and Noise Self-Model Broadcasts in an Evolving Multi-Agent Network

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Abstract. Many networks respond to measurements of their own state: markets to indices, organizations to metrics, platforms to trending signals. We present a controlled laboratory for this reflexive loop: an evolving agent population forms an adaptive network whose macrostate is compressed into a self-model and broadcast back — accurately, distorted, or as matched-bandwidth noise. Across 927 deterministic runs with paired seeds and pre-specified outcomes: measurement without broadcast is causally inert; under corrective responses false self-models pull the macrostate toward themselves far more strongly than matched noise (paired rank-biserial $r = 0.84$–0.88), and realistically structured descriptions steer strongest — a replayed genuine self-model from another run out-steers both; which lies come true depends on the response regime, an alarmist lie flipping from self-defeating under corrective responses to strongly self-fulfilling under conformist ones; and heritable attention to the broadcast declines across generations with its unreliability: populations fed false self-models evolve to ignore their observer.

Keywords: adaptive networks, agent-based modeling, reflexivity, self-fulfilling dynamics, evolution of trust, network resilience

1 Introduction

Adaptive networks — graphs whose topology coevolves with the dynamical states of their nodes — are a central object of network science [1, 2]. A distinctive subclass has received far less controlled attention: networks that respond to *descriptions of themselves*. Financial markets move on published indices computed from their own trades [3]; universities restructure around rankings derived from their own behavior [4]; recommendation platforms amplify items *because* an internal measurement declared them trending. In each case a macroscopic measurement of the system is compressed, published, and re-enters the system as an input to microscopic decisions — a loop Merton called the self-fulfilling prophecy [5] and whose measure-distorting variant is known as Goodhart's law [6].

Empirical study of this loop is confounded: in real systems one can rarely withhold the measurement, falsify it in controlled directions, or replay history

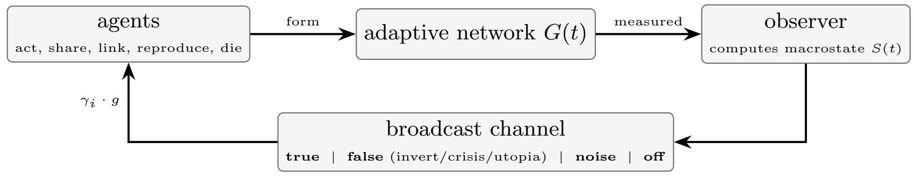

Fig. 1. The reflexive loop and its corruption points. Local actions generate the network; the observer compresses it into a self-model; the broadcast channel returns it truthfully, distorted, as noise, or not at all. Responses are scaled by heritable sensitivity γ_i and global gain g.

under identical randomness. We therefore built a minimal simulated world in which the loop can be opened, closed, and corrupted at will, where every run is exactly reproducible from a configuration and seed, and where agents are deliberately simple — transparent trait vectors rather than learned policies — so every pathway from broadcast to response is inspectable.

Our contributions are: (1) an open, deterministic experimental platform coupling an evolving agent ecology, an adaptive interaction network, a global observer, and a controllable broadcast channel; (2) a six-condition causal design that separates the effect of *being measured* from the effect of *being told*, the effect of *self-information* from that of *stimulation* (via a matched-bandwidth noise control), and the effect of *realistic structure* from that of self-reference (via a replayed-self-model control); (3) evidence, robust across feedback gains and two shock severities, that under corrective response rules false self-models steer the macrostate toward themselves far more strongly than matched noise — noise itself begins to steer at high gain, and steers as strongly as an inverted lie under conformist responses — with the steering direction set jointly by distortion type and response regime; and (4) we demonstrate, in an ecological agent-based setting, that *attention to a collective self-model is itself a heritable, evolvable trait* whose population mean declines with the self-model's unreliability. Figure 1 summarizes the design.

2 Related Work

Adaptive and temporal networks. Coevolution of state and topology is surveyed in [1, 2]; our model belongs to this family, with births, deaths and rewiring driven by agent decisions. Robustness of networks to random and targeted node removal is classical [7]; we use targeted hub removal as a standardized perturbation rather than an object of study. **Global coupling in physics.** Globally coupled oscillators and maps show that a mean field fed back to units reshapes collective dynamics [8]. Ours differs in two ways: the "mean field" is a multi-dimensional, potentially *false* description, and units are born, die, and evolve. **Reflexivity and reactive measurement.** Self-fulfilling prophecy [5], reflexivity in markets [3], and the reactivity of rankings and metrics [4, 6] motivate our false-broadcast conditions; we contribute a platform where these effects are dosable and ablatable. **Signaling and the evolution of trust.** In sender–receiver games, receivers can evolve to ignore unreliable signals [9]. Our distrust result is consonant with that theory but arises in an open ecological population where the signal is *self-referential* — a compressed description of the receiving population itself — and where attention to it is a continuous heritable trait under selection from an energy economy, not a payoff matrix. **Agent-based modeling.** Agent-based studies of collective opinion and cooperation are a mainstay of this community [16, 17]; our methodology follows the generative ABM tradition [10, 11], and the engine uses ideas from Mesa [12] and NetworkX [13]. Multi-agent reinforcement learning treats global state as an *architectural* aid [14]; we treat it as an *experimental variable* with controlled veracity. Populations of language-model agents are now deployed with shared dashboards describing their own collective state [15]; the present rule-based setting provides causally clean baselines for that regime.

3 The Model

3.1 Agents and ecology

Each agent i carries a heritable trait vector $\theta_i = (\mathrm{coop}, \mathrm{expl}, \mathrm{risk}, \mathrm{soc}, \mathrm{share}, \gamma_i) \in [0,1]^6$, where γ_i is its *global sensitivity* — how strongly broadcasts modulate its behavior. Agents hold energy; each step they pay metabolism and choose one action — harvest, share (costly transfer to a neighbor), connect, prune, reproduce, or idle — sampled with probabilities proportional to trait- and state-dependent propensities. The resource pool regenerates logistically, giving density-dependent competition. Reproduction requires an energy threshold and copies traits with Gaussian mutation ($\sigma = 0.05$); energy exhaustion or age causes death. Founding ages are staggered to avoid cohort artifacts.

3.2 Observer and broadcast conditions

Every $\Delta t = 10$ steps the observer computes the macrostate $S(t)$: population, fragmentation (1 minus largest-component share), degree- and betweenness-concentration, Freeman centralization [19], per-capita costly-helping rate, energy Gini, and turnover, plus mean trait values (logged, never broadcast). The broadcast $b(t)$ carries five of these components, each normalized to $[0,1]$: *fragmentation*, *centralization* (top-5% degree share), *cooperation*, *inequality*, and *turnover*. Six conditions define what agents receive (Table 2). Distorted broadcasts come in three modes: *invert* ($b_k = 1 - s_k$ componentwise), *crisis* (fixed alarming values: fragmentation 0.9, centralization 0.9, cooperation 0.1, inequality 0.9, turnover 0.9), and *utopia* (fixed reassuring values: 0.05, 0.2, 0.9, 0.1, 0.2).

3.3 Response model

Each step, agent i selects one action a with probability proportional to

$$w_i(a,t) \;=\; w_a\big(\theta_i, x_i(t)\big) \;+\; \gamma_i\, g\, f_a\big(b(t)\big), \tag{1}$$

Table 1. Model parameters (defaults; campaign overrides noted in the text).

initial population	150	observer interval Δt	10 steps
population cap	1200	feedback gain g	0.8 (swept 0.2–1.6)
initial edges per agent	3 (random)	mutation σ	0.05
initial energy	20	initial traits	mean 0.5, s.d. 0.15
resource capacity K	6000	reproduction threshold / cost	30 / 16
base regeneration	60/step	min. reproduction age	15 steps
regeneration rate r	0.12	lifespan (mean ± s.d.)	400 ± 80 steps
harvest cap	2.5/step	action cost / share amount	0.25 / 3.0
metabolism	0.8/step	run length	3000–4000 steps
shock time t_{shock}	2000		

Table 2. Experimental conditions. All start from matched parameter distributions with paired random seeds.

Condition	Observer	Agents receive	Role
A — Local only	off	nothing	baseline
B — Observed, blind	on	nothing	measurement control
C — True feedback	on	accurate $S(t)$	main treatment
F — False feedback	on	distorted $S(t)$	reflexivity probe
N — Noise feedback	on	uniform random, matched ranges	stimulation control
R — Replayed self-model	on	another run's genuine $S(t)$	structure control

where $w_a(\theta_i, x_i)$ is the trait- and local-state-dependent base propensity, g is the global feedback gain, and f_a is the response rule. In the primary *corrective* regime agents repair reported deficits: $f_{\mathrm{connect}} = \max(0,\, b_{\mathrm{frag}} - 0.3)$, $f_{\mathrm{share}} = \max(0,\, 0.5 - b_{\mathrm{coop}}) + 0.5\max(0,\, b_{\mathrm{ineq}} - 0.5)$, $f_{\mathrm{harvest}} = \max(0,\, b_{\mathrm{turn}} - 0.5)$, and $f_{\mathrm{prune}} = 0.3\max(0,\, b_{\mathrm{cent}} - 0.6)$. In the *conformist* regime agents imitate the reported norm instead: $f_{\mathrm{share}} = b_{\mathrm{coop}}$, $f_{\mathrm{prune}} = 0.5\, b_{\mathrm{frag}}$, $f_{\mathrm{connect}} = \max(0,\, 0.6 - b_{\mathrm{frag}})$, and $f_{\mathrm{harvest}} = b_{\mathrm{ineq}}$. Two further mechanisms act on *target selection* rather than action choice, in both regimes: while reported centralization exceeds 0.6, a pruning agent cuts its highest-degree neighbor (ties by id) with probability γ_i, else a uniform neighbor; and new links attach to a neighbor-of-a-neighbor with probability 0.7 when available (else a uniform non-neighbor) — an explicit triadic-closure tendency contributing to the clustering in Sect. 3.4. All rule constants are fixed across all experiments; Table 1 lists the model parameters.

3.4 Emergent network structure

The interaction networks that emerge are dense, clustered, and heterogeneous: at $t = 2{,}000$ the giant component has short characteristic path lengths ($L \approx 1.9$–2.5), high clustering (1.8–2.4× an Erdős–Rényi null of matched size and density), and a broad degree spread (coefficient of variation 0.6–0.75; maximum degree 2–2.5× the mean). Against the stricter degree-preserving (configuration-model) null, however, most of the clustering excess is accounted for by the degree sequence (observed/null ratios 1.0–1.6 across seeds and conditions), so we characterize the substrate simply as a small, dense, degree-heterogeneous network rather than claiming a small-world regime. The point relevant to what follows is that feedback acts on a heterogeneous, clustered graph rather than a homogeneous mixing population.

3.5 Reproducibility

The simulation is deterministic given (configuration, seed): one PRNG drives all choices, agents act in sorted order, and a state hash certifies replay; identical hashes were verified across operating systems and machines. All code, configurations, and analysis scripts are public.

4 Experimental Design

The standardized shock removes, in a single step at $t_{\text{shock}} = 2000$ (all campaigns), the highest-degree living agents (ties broken by agent id): a fixed 10 hubs in the mild variant, the top $\max(1, \lfloor 0.4\,|V_{\text{live}}|\rfloor)$ in the harsh variant. Four outcomes were fixed before the first campaign: (1) post-shock recovery time — the first $t > t_{\text{shock}}$ at which the living population regains 90% of its pre-shock baseline, defined as its mean over $[t_{\text{shock}}-500,\, t_{\text{shock}})$; (2) mean post-shock fragmentation; (3) per-capita costly-helping rate; (4) *story pull*: with $s_k(t)$ the value of macrostate component k at observer tick t, $b_k(t)$ the broadcast value, and $\varepsilon = 0.02$,

$$P \;=\; \Big\langle\, \operatorname{sign}\big(b_k(t) - s_k(t)\big)\,\big(s_k(t{+}1) - s_k(t)\big)\,\Big\rangle_{(t,k)\,:\,|b_k(t)-s_k(t)|>\varepsilon}, \tag{2}$$

averaged with equal weight over all qualifying (tick, component) pairs across the five broadcast components; $P > 0$ means the macrostate moves toward the broadcast where the two differ, and its magnitude is bounded by the largest per-tick component change. The threshold ε makes P well defined under false and noise broadcasts (and undefined for C, whose broadcast equals the current state); results are insensitive to its value — for $\varepsilon \in \{0.01, 0.02, 0.05\}$ the paired F-vs-N effect is $r = 0.88, 0.88, 0.86$ (all $p < 6\times10^{-6}$) and the steering hierarchy reported below is unchanged. Campaigns: a *flagship* run (conditions A/B/C/F_{invert}/N $\times$ 50 paired seeds, mild shock), a *harsh-shock* replication (removal of 40% of hubs, $\times$30 seeds), and a *sensitivity sweep* (feedback gain $g \in \{0.2, 0.8, 1.6\} \times \{$A, C, F_{invert}, F_{crisis}, F_{utopia}, N$\} \times 20$ seeds), a *scale check* re-running the core conditions with doubled population and resource capacity, a *response-regime campaign* repeating all broadcast conditions with conformist agents (15 seeds per cell), and a *replay control* (30 receiving runs paired with the harsh-shock seed list, plus 15 auxiliary source runs that generate the replayed trajectories): $250 + 150 + 360 + 90 + 32 + 30 + 15 = 927$ runs executed in total, each of 3,000–4,000 steps. Because conditions share seed lists, the design is paired; the primary analysis is a Wilcoxon signed-rank test on per-seed differences, reported with the

Table 3. Primary paired comparisons (Wilcoxon signed-rank on per-seed differences; matched-pairs rank-biserial r; bootstrap 95% CI of the median difference). FL = flagship (50 pairs), HS = harsh shock (30 pairs).

Outcome	Comparison	Δmedian [CI]	r	p
every outcome (FL, HS)	B vs A	all differences $\equiv 0$	—	—
story pull (FL)	F vs N	$+0.0055$ $[0.0033, 0.0064]$	0.84	1.1×10^{-8}
story pull (HS)	F vs N	$+0.0065$ $[0.0043, 0.0078]$	0.88	3.2×10^{-6}
story pull (HS)	R vs N	$+0.0212$ $[0.0181, 0.0243]$	1.00	1.9×10^{-9}
cooperation (FL)	C vs A	$+0.200$ $[0.172, 0.239]$	1.00	8.9×10^{-15}
cooperation (HS)	C vs A	$+0.209$ $[0.170, 0.248]$	0.99	9.3×10^{-9}
fragmentation (FL)	F vs A	-0.017 $[-0.034, -0.006]$	-0.53	9.5×10^{-4}
$\Delta\bar{\gamma}$ (HS)	F vs A	-0.336 $[-0.456, -0.281]$	-1.00	1.9×10^{-9}
$\Delta\bar{\gamma}$ (HS)	N vs A	-0.257 $[-0.333, -0.218]$	-0.98	1.9×10^{-8}
recovery time (FL, HS)	C, F, N vs A	Δmedian $= 0$	$\lvert r\rvert < 0.5$	> 0.22

matched-pairs rank-biserial correlation r and a bootstrap 95% CI on the median difference. Unpaired Mann–Whitney U with Cliff's δ is retained as a robustness check and agrees with the paired analysis throughout; all other logged measures are exploratory.

5 Results

Table 3 summarizes the pre-specified comparisons; the following subsections unpack them.

5.1 Measurement alone is causally inert

Under paired seeds, condition B is trajectory-identical to A: every per-seed difference is exactly zero, on every outcome, in both the flagship and harsh-shock campaigns. The instrument acquires causal power only when it broadcasts — a validation that the design isolates the channel of interest.

5.2 A false self-model pulls the macrostate toward itself

Except where stated otherwise, all results concern the primary corrective regime; Sect. 5.4 shows how they change under conformity. Story pull (Eq. 2) under the inverted false broadcast exceeds the noise control at every gain tested (paired: flagship Δmedian $+0.0055$, $r = 0.84$, $p \approx 1.1 \times 10^{-8}$; harsh shock $+0.0065$, $r = 0.88$, $p \approx 3.2 \times 10^{-6}$). Tested directly against zero (harsh shock; median P with bootstrap 95% CI of the median; one-sample Wilcoxon signed-rank), the pull is positive in every broadcast condition: F median 0.0078 $[0.0066, 0.0100]$; R (below) 0.0241 $[0.0197, 0.0265]$ (both $p \approx 1.9 \times 10^{-9}$); N 0.0021 $[0.0009, 0.0038]$ ($p \approx 10^{-3}$). Decomposed by component, the aggregate pull is carried chiefly by the turnover (≈ 0.033) and cooperation (≈ 0.005) components; the structural components are net-neutral, because the corrective over-connection documented below cancels drift toward the reported high fragmentation. Relative to noise the effect is strongest at *low* gain (ratio ≈ 7 at $g = 0.2$): a quiet, systematic lie out-steers loud noise, while at high gain noise itself begins to steer (Fig. 2, middle). Spillovers are substantial: the inverted broadcast reduces post-shock fragmentation relative to baseline (flagship paired Δmedian -0.017, $r = -0.53$, $p \approx 1 \times 10^{-3}$; directionally consistent but not significant in the harsh-shock replication, $p = 0.28$) because agents persistently told the network is fragmenting over-connect it (Fig. 3).

Realistic structure grades the steering channel. A stronger structured control replaces the broadcast with a *replayed genuine self-model*: the recorded $S(t)$ trajectory of an observed-but-blind run with a different seed (condition R; 30 receiving runs sharing the harsh-shock seed list, each replaying one of 15 source trajectories) — realistic marginals and temporal coherence, no self-reference. The resulting steering hierarchy is R > F > N (medians 0.024, 0.008, 0.002; paired over the 30 shared seeds, R vs N and F vs R both $|r| = 1.00$, $p \approx 1.9 \times 10^{-9}$). The result does not depend on the reuse of source trajectories: restricted to one receiver per source ($n = 15$ independent trajectories), R exceeds F and N in every pair ($r = 1.00$, $p = 6.1 \times 10^{-5}$ for both), and a source-clustered bootstrap over the full sample gives mean differences R$-$F $+0.0148$ $[0.0121, 0.0173]$ and R$-$N $+0.0207$ $[0.0178, 0.0235]$. Nor is it explained by how often responses fire: mean discrepancy $\langle |b_k - s_k| \rangle$ is 0.64 (F), 0.37 (N), and only 0.10 (R), and F holds four of five corrective rules active on $\geq 90\%$ of ticks against a mostly quiescent profile under R — R steers most while activating least. R is of course also a false description of the receiving network — a *realistically structured* one, true of a sibling world. Steering power is therefore graded by the realism of the description's structure, and the channel responds to that structure, not to persistence or stimulation volume alone.

Structural consequences. Broadcasts densify and flatten the network. Relative to A (paired, harsh shock, post-transient medians), mean degree rises by $+1.1$ under truth ($r = 0.49$) and by $+3.8/+4.2$ under false/noise broadcasts ($r = 0.80/0.79$), while Freeman centralization falls (F vs A: -0.026, $r = -0.74$) and betweenness concentration falls (F: -0.007, $r = -0.67$; N similar): the fed-back network becomes denser and less hub-dominated than the silent one.

5.3 The type of lie sets the direction

The three distortion modes produce qualitatively different regimes (Fig. 2). The *inverted* self-model, always reporting the mirror of reality, generates the strongest pull: it keeps every corrective response permanently active and thereby parasitizes the population's repair machinery. The *crisis* self-model is self-defeating for structure — corrections push fragmentation away from the alarmist report — yet its permanent cooperation-deficit signal mobilizes the highest costly-helping rates observed (0.36–0.45). The *utopia* self-model is behaviorally inert: because

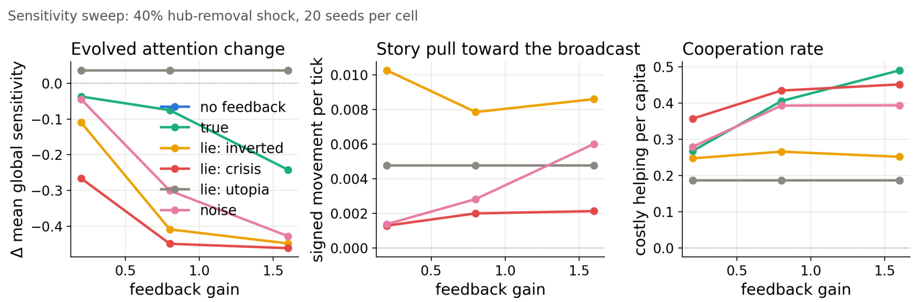

Fig. 2. Sensitivity sweep (40% hub-removal shock; 20 seeds per cell; medians). Left: evolved change in mean global sensitivity γ — attention to the self-model declines in proportion to broadcast unreliability and gain; the flattering (utopia) lie tracks the no-feedback baseline exactly. Middle: story pull toward the broadcast. Right: cooperation rate.

responses fire only on reported deficits, a flattering broadcast triggers nothing and is indistinguishable from disconnecting the channel, at every gain. Under purely corrective response rules, flattery equals silence. We note the caveat here rather than in the discussion: the first-order *directions* of these responses follow from the corrective rules (Eq. 1) by construction and are not claimed as discoveries; what is not built in is the magnitude asymmetry between distortion types, the spillovers to non-targeted structure, and the evolutionary consequences below.

5.4 The response regime decides which lies come true

Repeating all broadcast conditions with conformist agents reverses the taxonomy exactly as the mechanism predicts (Table 4). The *crisis* lie, self-defeating under correction, becomes strongly self-fulfilling under conformity: told the world is fragmenting, imitators cut ties and fragmentation rises an order of magnitude above baseline (0.400 vs 0.038). The *utopia* lie flips from behaviorally inert to benevolently self-fulfilling: "everyone cooperates" becomes true by imitation (cooperation 0.428, the highest observed in any condition). Under conformity noise steers as strongly as the inverted lie, since imitation amplifies any reported value. Which false self-models come true is therefore a property of the lie–response pair, not of the lie alone.

5.5 Populations fed lies evolve to stop listening

Mean heritable global sensitivity $\bar{\gamma}$ changes over $\sim$40 generations as an emergent, unprogrammed function of broadcast quality (Fig. 4). At $g = 0.8$ (harsh-shock campaign): no broadcast -0.03 (neutral drift), truth -0.10, noise -0.30 (paired $r = -0.98$), false -0.42 (paired Δmedian -0.336 $[-0.456, -0.281]$, $r = -1.00$, $p \approx 1.9 \times 10^{-9}$). The sweep reveals a dose–response: the decline under the false

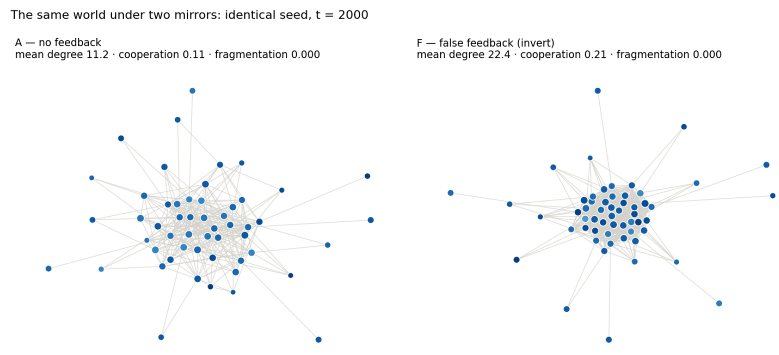

Fig. 3. The same world under two mirrors. Two runs from an identical seed at $t = 2000$: without feedback (left) and under the inverted false broadcast (right), which persistently reports fragmentation and thereby drives over-connection (mean degree 22.4 vs 11.2) and elevated costly helping. Node size encodes energy; color encodes generation.

Table 4. The same lie under two response regimes (medians; harsh shock, $g = 0.8$). Baseline (A): cooperation 0.18, fragmentation 0.038.

Lie	Corrective		Conformist	
	coop.	fragm.	coop.	fragm.
Invert	0.27	0.021	0.42	0.054
Crisis	0.36	0.025	0.26	**0.400**
Utopia	0.19 (inert)	0.037 (inert)	**0.428**	0.025

broadcast deepens from -0.11 ($g = 0.2$) to -0.45 ($g = 1.6$) — and at the highest gain even *accurate* feedback shows a marked decline (-0.24): high-impact feedback is disfavored even when true. The utopia lie, being consequence-free, produces no decline at all.

We state the claim at the level the data support: a systematic, dose-dependent shift in the frequency of a heritable trait under matched conditions, whose individual-level pathway is not a simple fecundity gradient — the within-run correlation between γ_i and offspring count is not negative under F ($\rho \approx +0.05$) — and not shock-mediated (F without the shock: -0.40 vs -0.39 with it; shock victims' mean γ exceeds the population mean by only $+0.03$). The pathway plausibly runs through lineage survival and energy budgets downstream of reproduction; identifying it precisely is left open.

5.6 Accurate feedback as distributed corrective feedback

True feedback roughly doubles costly helping (flagship: 0.397 vs 0.179; paired Δmedian $+0.200$ $[0.172, 0.239]$, $r = 1.00$, $p \approx 9 \times 10^{-15}$; replicated under harsh

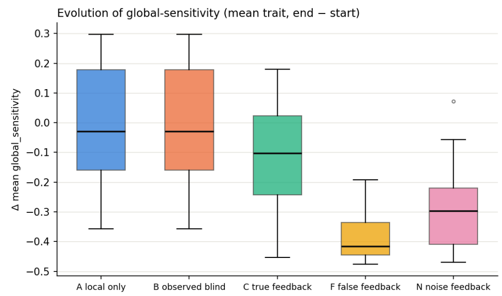

Fig. 4. Evolution of attention to the self-model (harsh-shock campaign, $g = 0.8$, 30 seeds per condition). Change in population-mean global sensitivity $\bar{\gamma}$, run start to end. The ordering (false < noise < true < none) emerged from the evolutionary dynamics alone; nothing in the rules references broadcast reliability.

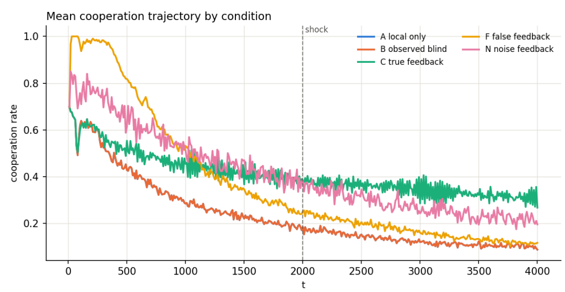

Fig. 5. Mean cooperation trajectories by condition (flagship campaign, 50 seeds per condition; dashed line marks the standardized shock). Conditions A and B coincide exactly. True and noise feedback sustain elevated cooperation; the inverted false broadcast sustains an intermediate level.

shock, $r = 0.99$), rising monotonically with gain ($0.27 \to 0.49$); Fig. 5 shows the sustained divergence of trajectories. The loop implements distributed corrective feedback: deficits reported at the population level are repaired by uncoordinated individual action.

5.7 A pre-specified null result

We detected no evidence of a feedback effect on post-shock recovery time in any condition or campaign, even with 40% of hubs removed (all $p > 0.12$; median $\approx$ 40 steps throughout). Within the regimes tested, resilience appears demographic — fast regrowth — rather than informational; we report this pre-specified null as observed rather than adjusting the protocol until it disappears.

5.8 Robustness to system scale

Re-running the core conditions (A, C, F_{invert}, N) with doubled initial population and proportionally scaled resources (steady-state population $\approx$ 120–130 vs. ≈ 60) preserves every effect under the paired analysis ($n = 8$ seed pairs per condition): cooperation under true feedback is 0.399 vs. 0.188 baseline (Δmedian $+0.151$ $[0.119, 0.247]$, $r = 1.00$, $p = 0.008$), story pull retains F vs. N superiority (Δmedian $+0.0046$ $[0.0022, 0.0070]$, $r = 0.89$, $p = 0.023$), and the distrust ordering is intact with F vs. A at Δmedian -0.299 $[-0.388, -0.212]$, $r = -1.00$, $p = 0.008$. The principal phenomena therefore persist under a $2\times$ increase in system scale.

6 Discussion and Limitations

As noted alongside the results, the first-order response directions are encoded in the rules; the findings live one level up: the magnitude and asymmetry of steering (graded by the realism of the description's structure; under corrective responses consistent lies $\gg$ matched noise, strongest at low gain), the spillovers to non-targeted structure, the regime-dependent taxonomy of lies (exploitative / mobilizing-but-self-defeating / inert under correction; reversed under conformity), and above all the evolutionary layer, in which the credibility of a collective self-model becomes an evolving property of the population. Limitations: a single ecology and parameter family (though effects are monotone across an $8\times$ gain range and stable under a $2\times$ scale change), fixed rule-based rather than learned response policies, and one network per run rather than interacting populations.

The results have a practical reading for engineered systems. Metric-driven infrastructure — autoscalers, observability pipelines, trending algorithms — constitutes exactly this loop; the false-broadcast conditions model telemetry corruption, and the utopia result suggests that under deficit-driven responses *flattering telemetry can be operationally equivalent to no telemetry* while raising no alarms. Documented episodes of measurement reshaping the measured system — from ranking reactivity in institutions [4] to the degradation of Google Flu Trends once its predictions entered the environment they forecast [18] — are natural empirical targets, and connecting the model's regimes to such datasets is the clearest path to empirical grounding. For multi-agent AI, where fleets of agents increasingly receive dashboards of their own collective state, the evolution-of-distrust result suggests that persistent misreporting does not merely misdirect a population — it degrades the control channel itself.

7 Conclusion

We introduced a reproducible causal laboratory for reflexive self-modeling in adaptive networks and used it to show that (i) being measured changes nothing while being told changes much; (ii) under corrective responses a consistent false self-description steers a collective toward itself far more effectively than noise

— with realistically structured descriptions steering strongest — while under conformist responses noise steers as strongly as an inverted lie; (iii) which lies come true is set by the lie–response pair, not the lie alone; and (iv) heritable attention to the collective self-model declines across generations in proportion to the self-model's unreliability and impact, making a population's trust in its own self-model an evolving quantity — one that unreliable observers spend down. Future work: continuity under total population turnover, learned response policies, and language-model agents.

Reproducibility. Code, configurations, all 912 stored run manifests plus the 15 replay-source trajectories, and one-command reproduction of every figure: `github.com/Ivan825/Project-ONE` at tag `cn2026-submission` — the exact state described here.

References

1. Gross, T., Blasius, B.: Adaptive coevolutionary networks: a review. J. R. Soc. Interface 5(20), 259–271 (2008)
2. Holme, P., Saramäki, J.: Temporal networks. Phys. Rep. 519(3), 97–125 (2012)
3. Soros, G.: Fallibility, reflexivity, and the human uncertainty principle. J. Econ. Methodol. 20(4), 309–329 (2013)
4. Espeland, W.N., Sauder, M.: Rankings and reactivity: how public measures recreate social worlds. Am. J. Sociol. 113(1), 1–40 (2007)
5. Merton, R.K.: The self-fulfilling prophecy. Antioch Rev. 8(2), 193–210 (1948)
6. Strathern, M.: 'Improving ratings': audit in the British university system. Eur. Rev. 5(3), 305–321 (1997)
7. Albert, R., Jeong, H., Barabási, A.-L.: Error and attack tolerance of complex networks. Nature 406, 378–382 (2000)
8. Kuramoto, Y.: Chemical Oscillations, Waves, and Turbulence. Springer (1984)
9. Skyrms, B.: Signals: Evolution, Learning, and Information. Oxford Univ. Press (2010)
10. Epstein, J.M., Axtell, R.: Growing Artificial Societies. Brookings/MIT Press (1996)
11. Bonabeau, E.: Agent-based modeling: methods and techniques for simulating human systems. PNAS 99(suppl. 3), 7280–7287 (2002)
12. Kazil, J., Masad, D., Crooks, A.: Utilizing Python for agent-based modeling: the Mesa framework. In: SBP-BRiMS. Springer (2020)
13. Hagberg, A., Schult, D., Swart, P.: Exploring network structure, dynamics, and function using NetworkX. In: SciPy (2008)
14. Foerster, J., Assael, Y.M., de Freitas, N., Whiteson, S.: Learning to communicate with deep multi-agent reinforcement learning. In: NeurIPS (2016)
15. Park, J.S., O'Brien, J., Cai, C.J., Morris, M.R., Liang, P., Bernstein, M.S.: Generative agents: interactive simulacra of human behavior. In: UIST (2023)
16. Castellano, C., Fortunato, S., Loreto, V.: Statistical physics of social dynamics. Rev. Mod. Phys. 81, 591–646 (2009)
17. Perc, M., Jordan, J.J., Rand, D.G., Wang, Z., Boccaletti, S., Szolnoki, A.: Statistical physics of human cooperation. Phys. Rep. 687, 1–51 (2017)
18. Lazer, D., Kennedy, R., King, G., Vespignani, A.: The parable of Google Flu: traps in big data analysis. Science 343, 1203–1205 (2014)
19. Freeman, L.C.: Centrality in social networks: conceptual clarification. Soc. Netw. 1(3), 215–239 (1978)